

Calculating Probabilities Using R

BIO210 Biostatistics

Extra Reading Material for Lecture 20

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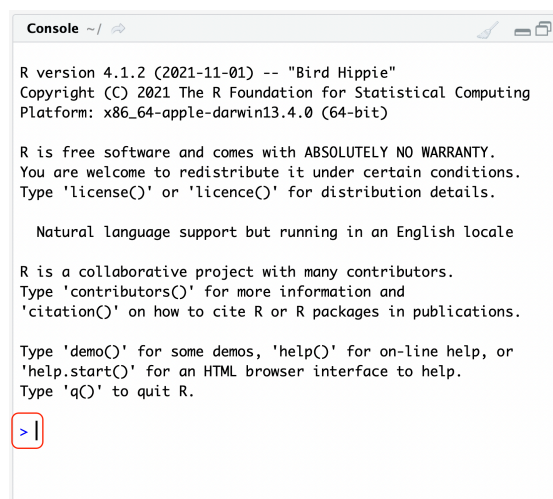
Spring, 2026

1 Calculating Probabilities Using R

As we gradually introduce more and more statistic concepts and tests, we will need to do quite a few complicated calculations. Some of them cannot be computed by hand or regular calculator; others can, but there are just too many steps and they are error-prone. Therefore, we introduce how to use the programming language **R** for the calculation of probabilities and related stuff. You don't really need to know how to write scripts in **R**. For now, you can just use it as a calculator.

Go to the SUSTech mirror of **R**: <https://mirrors.sustech.edu.cn/CRAN>. Download and install the program based on your operating system.

If you open the program, you will probably see a window like shown in the screenshot on the right-hand side. The “>” mark that I highlighted at the bottom is your “prompt”, where you can type commands. Like I said before, for now, you don't really need to know how to write scripts in **R**. You just use it as a probability calculator. Later on, we will see how to perform statistical tests.



```
Console ~/ 
R version 4.1.2 (2021-11-01) -- "Bird Hippie"
Copyright (C) 2021 The R Foundation for Statistical Computing
Platform: x86_64-apple-darwin13.4.0 (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

>|
```

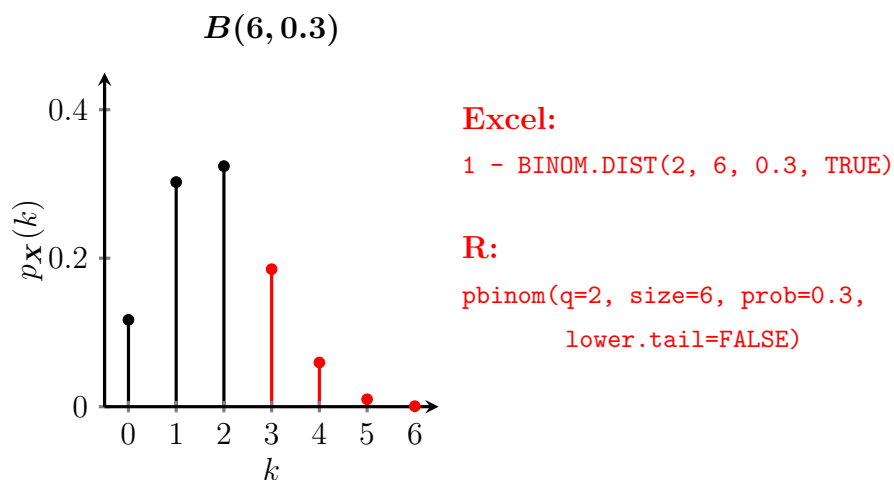
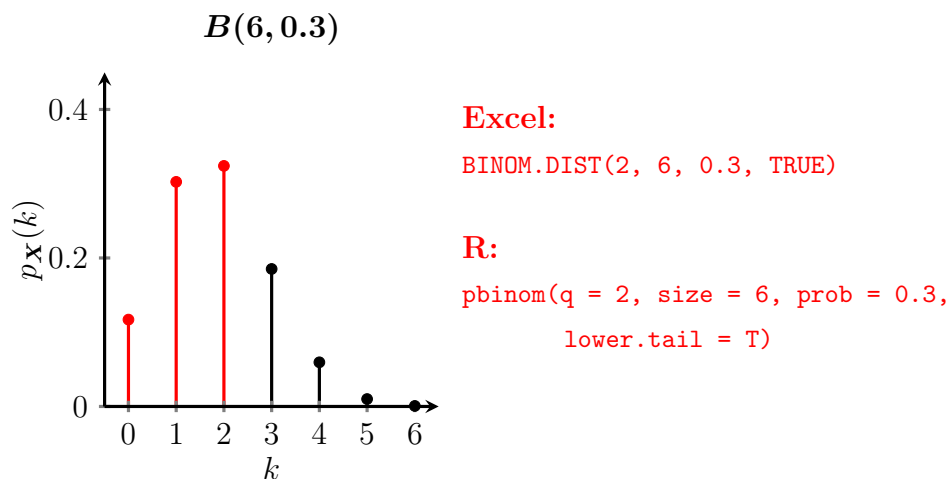
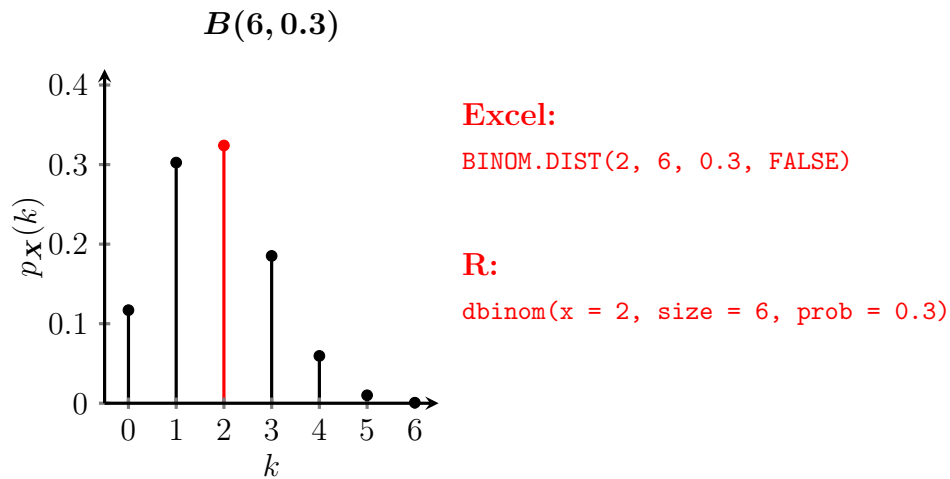
Since we already introduced how to calculate probability using **Excel**, we will use that as the comparison. Let's start with simple examples of three different distributions: $B(6, 0.3)$, $Pois(2)$ and $\mathcal{N}(1, 4)$. In **R**, if you want the cumulative probability, the function always starts with “p”; if you want the probability mass/density, the function always starts with “d”. For example, **pbinom**, **ppois**, and **pnorm** will give us the cumulative probability of the binomial, Poisson and normal distributions, respectively. They are like the ***.DIST** functions in **Excel** with the “cumulative” option set to “TRUE or 1”.

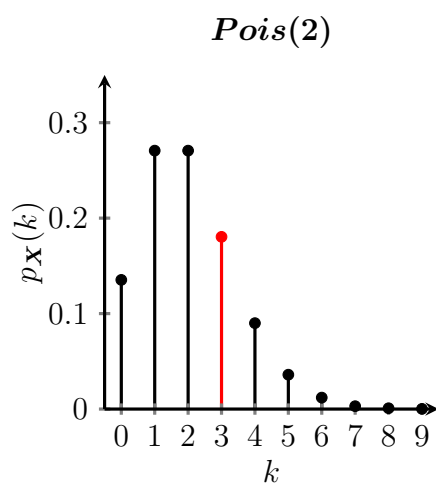
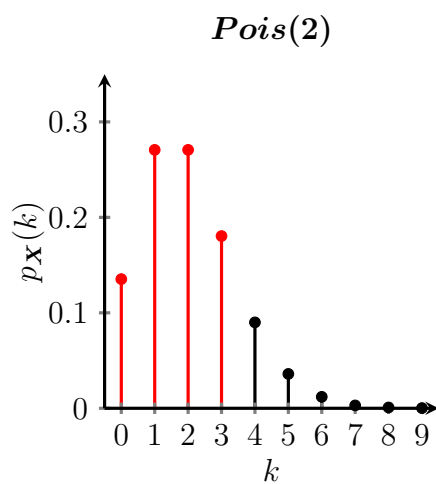
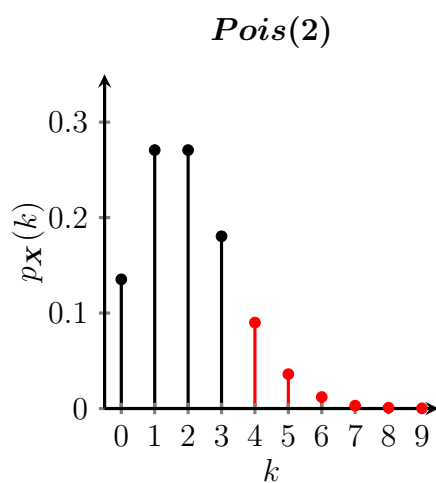
Similarly, the **dbinom**, **dpois** will give the exact probability of a binomial and Poisson distributions respectively, and **dnorm** will give you the probability density of a normal distribution. They are like the ***.DIST** functions in **Excel** with the “cumulative” option set to “FALSE or 0”.

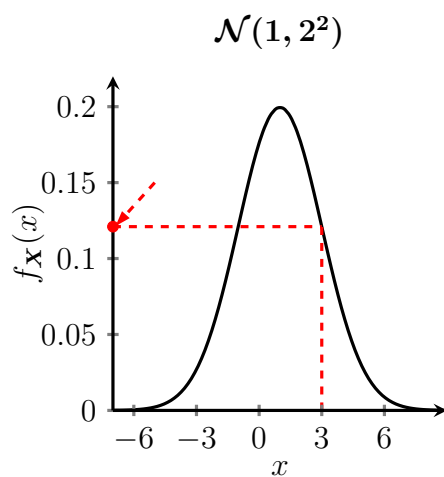
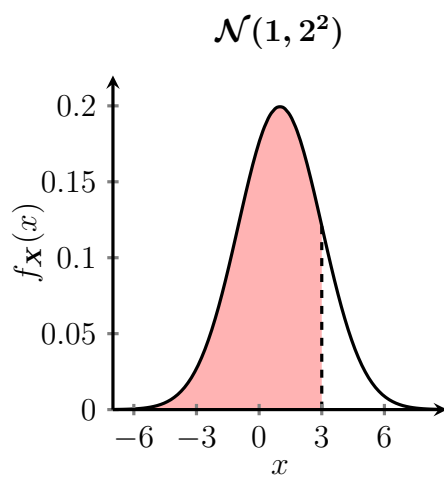
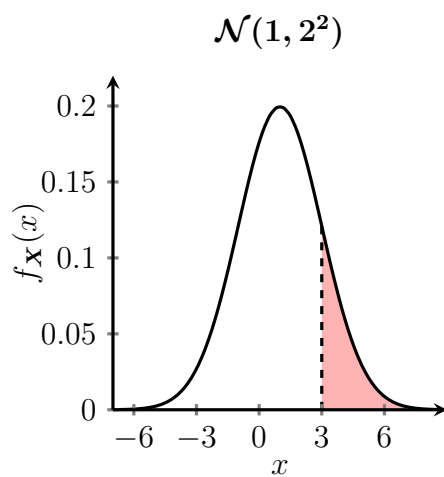
If you want to do the opposite, that is, provide the cumulative probability and

return the value of the random variable, the function always starts with “q”, like `qbinom`, `qpois` and `qnorm`. They are like the `*.INV` functions in **Excel**.

Just to give you some visual explanation of typical calculations:

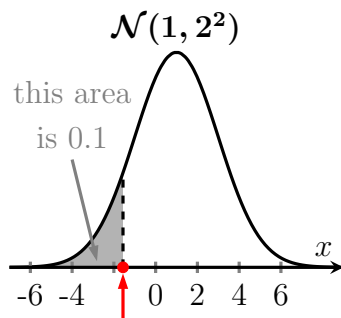


**Excel:**`POISSON.DIST(3, 2, FALSE)`**R:**`dpois(x = 3, lambda = 2)`**Excel:**`POISSON.DIST(3, 2, TRUE)`**R:**`ppois(q = 3, lambda = 2,
lower.tail = TRUE)`**Excel:**`1 - POISSON.DIST(3, 2, TRUE)`**R:**`ppois(q = 3, lambda = 2,
lower.tail = FALSE)`

**Excel:**`NORM.DIST(3, 1, 2, FALSE)`**R:**`dnorm(x = 3, mean = 1, sd = 2)`**Excel:**`NORM.DIST(3, 1, 2, TRUE)`**R:**`pnorm(q = 3, mean = 1, sd = 2,
lower.tail = TRUE)`**Excel:**`1 - NORM.DIST(3, 1, 2, TRUE)`**R:**`pnorm(q = 3, mean = 1, sd = 2,
lower.tail = FALSE)`

2 Find x with certain probability in a normal distribution

Again, using the function from **Excel** as a comparison, visually:

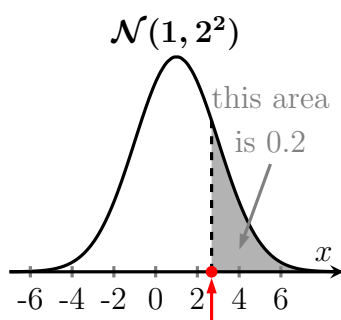


Excel:

`NORM.INV(0.1, 1, 2)`

R:

`qnorm(p = 0.1, mean = 1, sd = 2,
lower.tail = TRUE)`



Excel:

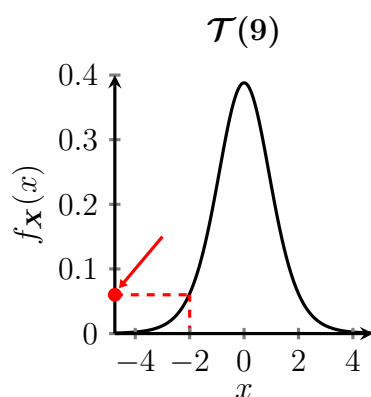
`NORM.INV(0.8, 1, 2)`

R:

`qnorm(p = 0.2, mean = 1, sd = 2,
lower.tail = FALSE)`

3 More Complicated Distributions

For more complicated distributions that we will encounter later in the course, the principle is exactly the same.

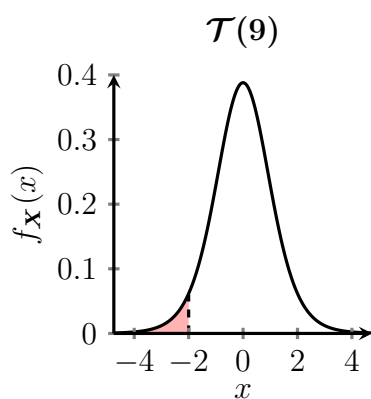
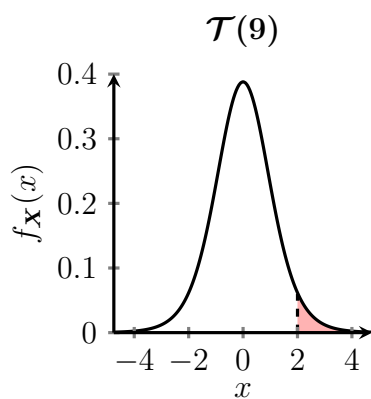
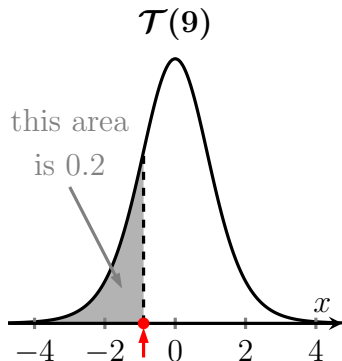
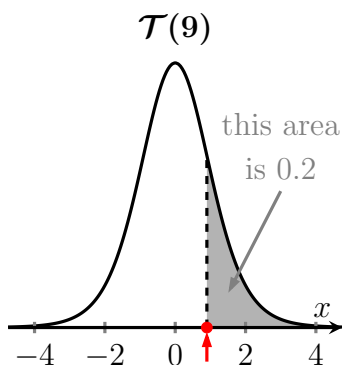


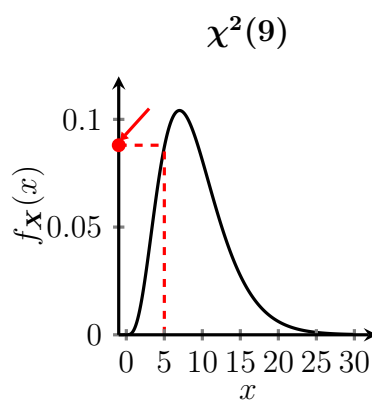
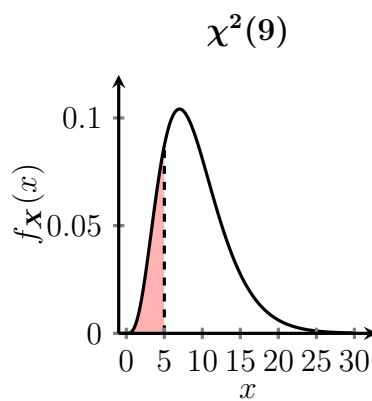
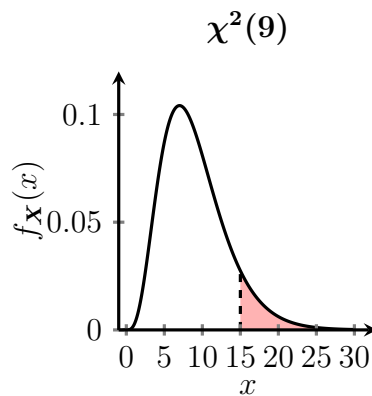
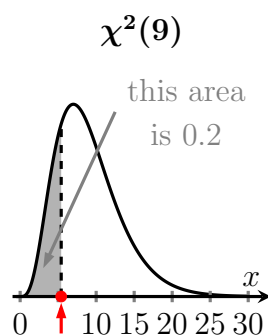
Excel:

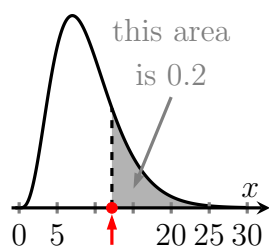
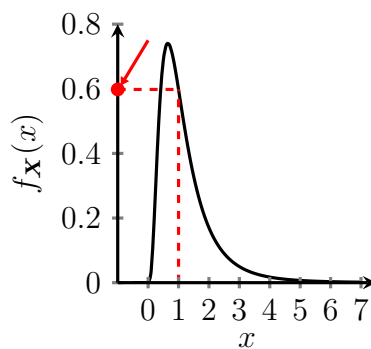
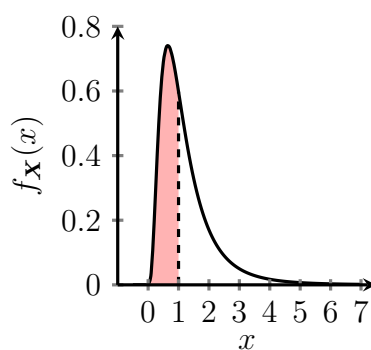
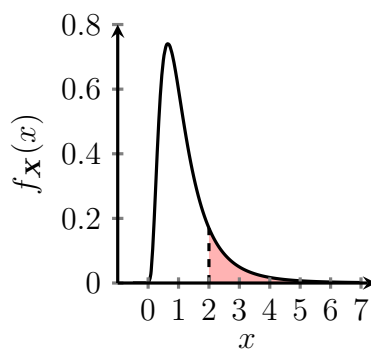
`T.DIST(-2, 9, FALSE)`

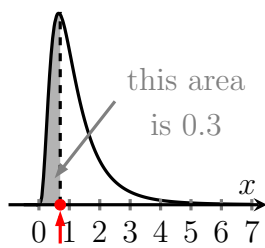
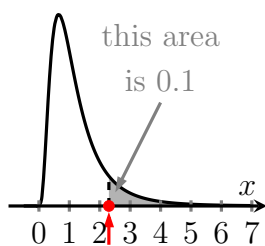
R:

`dt(x = -2, df = 9)`

**Excel:**`T.DIST(-2, 9, TRUE)`**R:**`pt(q = -2, df = 9, lower.tail = TRUE)`**Excel:**`1 - T.DIST(2, 9, TRUE)`**R:**`pt(q = -2, df = 9, lower.tail = FALSE)`**Excel:**`T.INV(0.2, 9)`**R:**`qt(p = 0.2, df = 9, lower.tail = TRUE)`**Excel:**`T.INV(0.8, 9)`**R:**`qt(p = 0.2, df = 9, lower.tail = FALSE)`

**Excel:**`CHISQ.DIST(5, 9, FALSE)`**R:**`dchisq(x = 5, df = 9)`**Excel:**`CHISQ.DIST(5, 9, TRUE)`**R:**`pchisq(q = 5, df = 9, lower.tail = TRUE)`**Excel:**`1 - CHISQ.DIST(15, 9, TRUE)`**R:**`pchisq(q = 15, df = 9, lower.tail = FALSE)`**Excel:**`CHISQ.INV(0.2, 9)`**R:**`qchisq(p = 0.2, df = 9, lower.tail = TRUE)`

$\chi^2(9)$ **Excel:**`CHISQ.INV(0.8, 9)`**R:**`qchisq(p = 0.2, df = 9, lower.tail = FALSE)` $\mathcal{F}(9, 10)$ **Excel:**`F.DIST(1, 9, 10, FALSE)`**R:**`df(x = 1, df1 = 9, df2 = 10)` $\mathcal{F}(9, 10)$ **Excel:**`F.DIST(1, 9, 10, TRUE)`**R:**`pf(q = 1, df1 = 9, df2 = 10,
lower.tail = TRUE)` $\mathcal{F}(9, 10)$ **Excel:**`1 - F.DIST(2, 9, 10, TRUE)`**R:**`pf(q = 2, df1 = 9, df2 = 10,
lower.tail = FALSE)`

$\mathcal{F}(9, 10)$ **Excel:**`F.INV(0.3, 9, 10)`**R:**`qf(p = 0.3, df1 = 9, df2 = 10, lower.tail = TRUE)` $\mathcal{F}(9, 10)$ **Excel:**`F.INV(0.9, 9, 10)`**R:**`qf(p = 0.1, df1 = 9, df2 = 10, lower.tail = FALSE)`